A TYPE-2 SUBCASE OF THE DUAL ADAPTED SQUARE-FUNCTION ESTIMATE

FA/BANACH AUTOMATIC DISCOVERY RUN

Status

This is a partial result adjacent to the lattice-assumption question in T. P. Hytönen and A. V. Vähäkangas, *The local non-homogeneous Tb theorem for vector-valued functions*, arXiv:1201.0648. It proves the adjoint adapted square-function estimate of source Theorem 4.16 under UMD plus Rademacher type 2, rather than under the source's UMD function-lattice hypothesis. It does not settle the corresponding estimate for arbitrary UMD spaces.

Source Question

The source explains that the function-lattice assumption is used at one specific square-function step and explicitly asks for possible removal of that assumption:

> **Estimate for the adjoints** $(D_k^a)^*$**.** Here we prove a norm estimate under the UMD function lattice assumption. The need for this assumption was somewhat unexpected to us, but with our present techniques, we were unable to avoid it. Proving (or disproving) the dual square-function estimate in the absence of the lattice assumption would be an interesting question for a deeper understanding of the vector-valued theory.
>
> 4.16. **Theorem.** *Let X be a* UMD *function lattice and $1 < p < \infty$. Then*

In the notation of the source, the target estimate is

$$\Big\|\sum_{k\in\mathbb{Z}} \varepsilon_k (D_k^a)^* f\Big\|_{L^p(\mathbb{R}^N\times\Omega,\mu\otimes\mathbf{P};X)} \lesssim \|f\|_{L^p(\mathbb{R}^N,\mu;X)}.$$

Proof Intuition

The lattice proof of the source uses pointwise moduli:

$$\Big\|\sum \varepsilon_k \xi_k\Big\|_{L^p_\varepsilon(X)} \simeq \Big\|\Big(\sum |\xi_k|^2\Big)^{1/2}\Big\|_X .$$

This is unavailable in a general Banach space. However, if X has Rademacher type 2, then the random sum is still controlled pointwise by the scalar square

Date: 2026-07-18.

function of the norms:

$$\left(\mathbb{E}_\varepsilon \left\|\sum \varepsilon_k \xi_k\right\|_X^p\right)^{1/p} \lesssim_{p,X} \left(\sum \|\xi_k\|_X^2\right)^{1/2}.$$

After Jensen's inequality gives $\|E_k(c_k f)\|_X \leq E_k(\|f\|_X)$, the remaining bound is just the scalar Carleson embedding theorem already present in the source. This replaces the only lattice-only ingredient in the proof of Theorem 4.16.

Lemma 1 (type-2 Carleson replacement)**.** *Let X have Rademacher type 2, let $1 < p < \infty$, and let $(E_k)_{k\in\mathbb{Z}}$ be the dyadic conditional expectations used in the source. Let $(d_k)_{k\in\mathbb{Z}}$ be scalar functions with $d_k = E_k d_k$ and finite source Carleson norm $\|\{d_k\}\|_{\mathrm{Car}^1(\mathcal{D})}$. Let $(c_k)_{k\in\mathbb{Z}}$ be scalar functions with $\|c_k\|_\infty \leq 1$. Then*

$$\left\|\sum_{k\in\mathbb{Z}} \varepsilon_k d_k E_k(c_k f)\right\|_{L^p(\mathbb{R}^N\times\Omega,\mu\otimes\mathbf{P};X)} \lesssim_{p,X} \|\{d_k\}\|_{\mathrm{Car}^1(\mathcal{D})}\|f\|_{L^p(\mathbb{R}^N,\mu;X)}.$$

The same conclusion holds with E_{k-1} in place of E_k, after reindexing d_k.

Proof. It is enough to prove the estimate for finite sums. By Kahane–Khintchine and Rademacher type 2, for each x,

$$\left(\mathbb{E}_\varepsilon \left\|\sum_k \varepsilon_k d_k(x) E_k(c_k f)(x)\right\|_X^p\right)^{1/p} \lesssim_{p,X} \left(\sum_k |d_k(x)|^2 \|E_k(c_k f)(x)\|_X^2\right)^{1/2}.$$

Since $\|c_k\|_\infty \leq 1$, conditional Jensen gives

$$\|E_k(c_k f)(x)\|_X \leq E_k(\|f\|_X)(x).$$

Therefore the left hand side in the lemma is bounded by

$$C_{p,X} \left\|\left(\sum_k |d_k|^2 (E_k\|f\|_X)^2\right)^{1/2}\right\|_{L^p(\mu)}.$$

By the scalar Kahane–Khintchine inequality this scalar square function is equivalent to

$$\left\|\sum_k \varepsilon_k d_k E_k(\|f\|_X)\right\|_{L^p(\mathbb{R}^N\times\Omega,\mu\otimes\mathbf{P})}.$$

The scalar case of the source Carleson embedding theorem now gives the desired bound, because $\|\|f\|_X\|_{L^p(\mu)} = \|f\|_{L^p(\mu;X)}$. □

Theorem 1 (type-2 adjoint square-function estimate)**.** *Let X be a* UMD *Banach space of Rademacher type 2, and let $1 < p < \infty$. For every $f \in L^p(\mathbb{R}^N, \mu; X)$, the adapted martingale differences of the source satisfy*

$$\left\|\sum_{k\in\mathbb{Z}} \varepsilon_k (D_k^a)^* f\right\|_{L^p(\mathbb{R}^N\times\Omega,\mu\otimes\mathbf{P};X)} \lesssim_{p,X,\delta} \|f\|_{L^p(\mathbb{R}^N,\mu;X)}.$$

Proof. We follow the proof of source Theorem 4.16 and replace only the lattice embedding Proposition 4.18.

The preliminary estimate for the non-adjoint adapted martingale differences, source Theorem 4.1, already holds for all UMD spaces. The algebraic representation of $(D_k^a)^* f$, the stopping-time identities, and the Carleson estimates for the scalar coefficient families

$$d_k = E_k b_k^a - E_{k-1} b_{k-1}^a, \qquad \chi_{k-1} = 1_{\{b_{k-1}^a \neq b_k^a\}},$$

are scalar and do not use the lattice structure of X.

Whenever the source invokes Proposition 4.18, the term has the form

$$\sum_k \varepsilon_k d_k E_k(c_k f) \quad \text{or} \quad \sum_k \varepsilon_k d_k E_{k-1}(c_k f),$$

where the c_k are scalar functions bounded in L^∞ by a constant depending only on the accretivity parameter δ, and the coefficient sequence d_k has uniformly bounded source Car^1 norm. After absorbing the harmless scalar bound on c_k, Lemma 1 gives the same estimate as Proposition 4.18 for these terms.

The remaining part of the source proof is unchanged: denominators $E_k b_k^a$ are removed by the contraction principle using the accretivity lower bound, the martingale-transform steps use the UMD property of X, and the final reindexing of the difficult term $\sum_k \varepsilon_k D_k(b_k^a f)$ is purely dyadic. Thus every term in the source decomposition is bounded by $\lesssim_{p,X,\delta} \|f\|_{L^p(\mu;X)}$, proving the displayed estimate. □

Scope and Novelty Check

Theorem 1 shows that the function-lattice structure is not necessary for the adjoint square-function estimate itself: Rademacher type 2 supplies enough square-function domination. The theorem remains a partial answer because it does not cover UMD spaces of type $r < 2$, and it does not claim the full operator-valued local Tb theorem under only UMD.

Cheap run-index searches for arXiv:1201.0648, the source title, the dual square-function estimate, and the type-2 Carleson replacement found no prior packet. Web searches on 2026-07-18 for the exact source title, the exact phrase "Proving (or disproving) the dual square-function estimate", and variants involving "type 2", "Carleson embedding", and "$E_k(c_k f)$" found the source paper and broader RMF/UMD literature, but no exact indexed settlement of this subcase.

References

References

[1] T. P. Hytönen and A. V. Vähäkangas, *The local non-homogeneous Tb theorem for vector-valued functions*, Glasg. Math. J. 57 (2015), no. 1, 17–82. arXiv:1201.0648.

[2] J. Bourgain, *Some remarks on Banach spaces in which martingale difference sequences are unconditional*, Ark. Mat. 21 (1983), no. 2, 163–168.

[3] T. Hytönen, A. McIntosh, and P. Portal, *Kato's square root problem in Banach spaces*, J. Funct. Anal. 254 (2008), no. 3, 675–726.

[4] T. Hytönen and M. Kemppainen, *On the relation of Carleson's embedding and the maximal theorem in the context of Banach space geometry*, Math. Scand. 109 (2011), no. 2, 269–284.

Human-Review Recommendation

Review as a likely valid partial result. The main point to verify is that every use of source Proposition 4.18 in the proof of source Theorem 4.16 has the coefficient form covered by Lemma 1; the source itself supplies the needed scalar Carleson bounds for those coefficients.